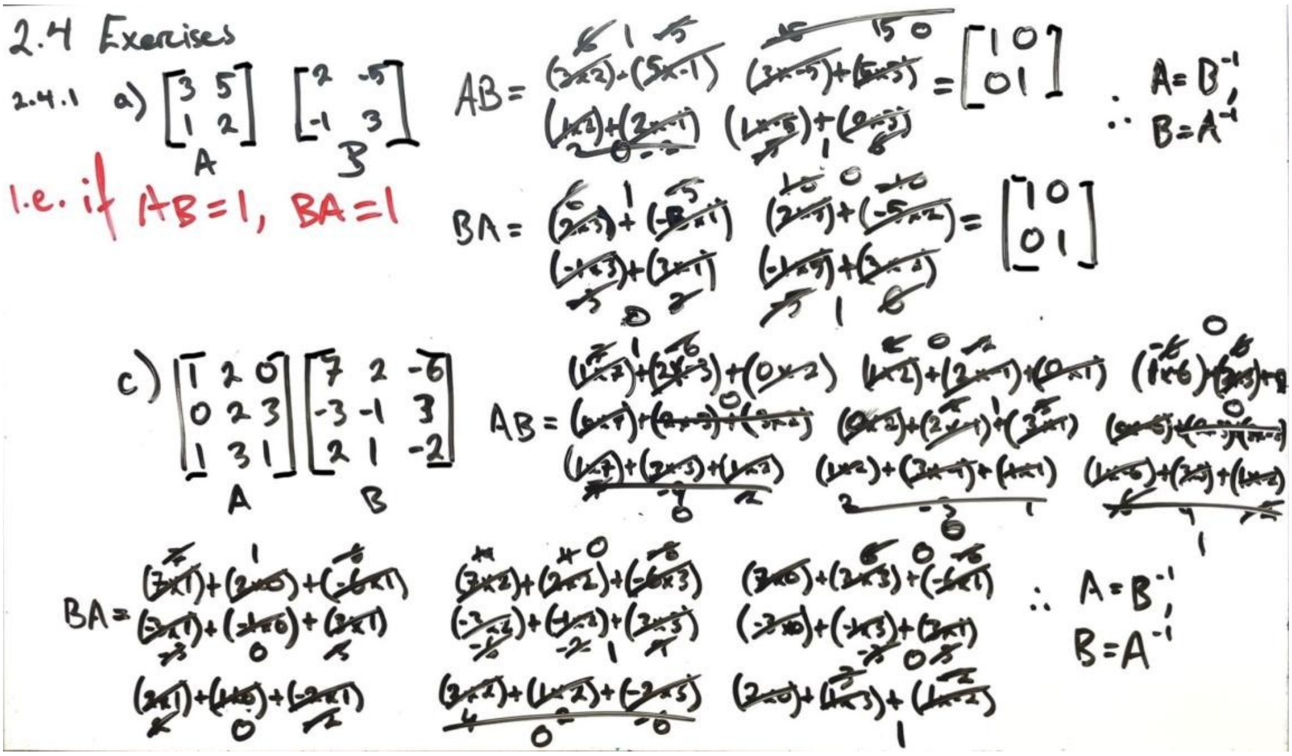


2.4 Exercises

2.4.1 a) $\underset{A}{\begin{bmatrix}3&5\\1&2\end{bmatrix}}$ $\underset{B}{\begin{bmatrix}2&-5\\-1&3\end{bmatrix}}$

$$AB=\begin{matrix}\overset{\cancel{6}\ 1\ \cancel{-5}}{\cancel{(3\times2)}+\cancel{(5\times-1)}} & \overset{\cancel{-15}\ 15\ 0}{\cancel{(3\times-5)}+\cancel{(5\times3)}}\\ \underset{\cancel{2}\ 0\ \cancel{-2}}{\cancel{(1\times2)}+\cancel{(2\times-1)}} & \underset{\cancel{-5}\ 1\ \cancel{6}}{\cancel{(1\times-5)}+\cancel{(2\times3)}}\end{matrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}$$

$\therefore A=B^{-1}, B=A^{-1}$

i.e. if $AB=1$, $BA=1$

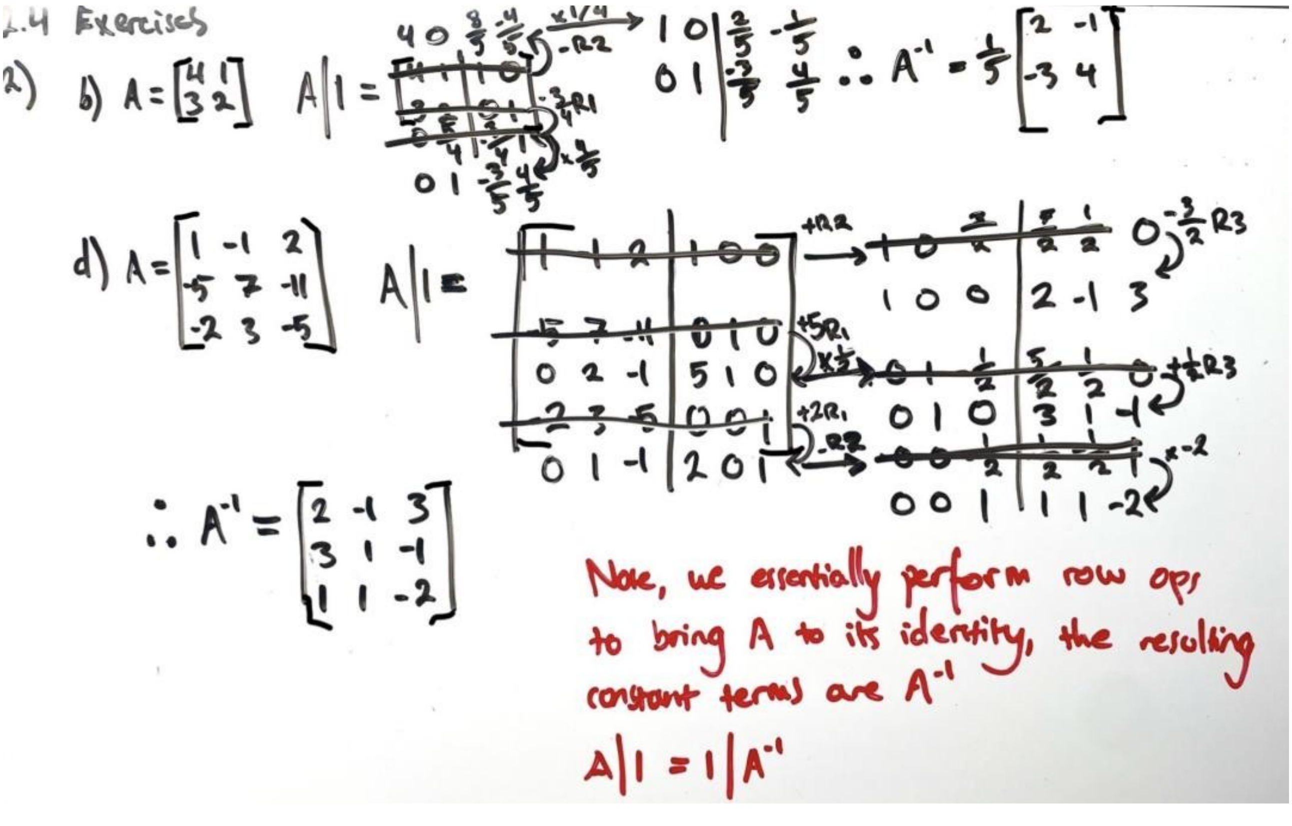
$$BA=\begin{matrix}\overset{\cancel{6}\ 1\ \cancel{-5}}{\cancel{(2\times3)}+\cancel{(-5\times1)}} & \overset{\cancel{10}\ 0\ \cancel{-10}}{\cancel{(2\times5)}+\cancel{(-5\times2)}}\\ \underset{\cancel{-3}\ 0\ \cancel{2}}{\cancel{(-1\times3)}+\cancel{(3\times1)}} & \underset{\cancel{-5}\ 1\ \cancel{6}}{\cancel{(-1\times5)}+\cancel{(3\times2)}}\end{matrix}=\begin{bmatrix}1&0\\0&1\end{bmatrix}$$

c) $\underset{A}{\begin{bmatrix}1&2&0\\0&2&3\\1&3&1\end{bmatrix}}$ $\underset{B}{\begin{bmatrix}7&2&-6\\-3&-1&3\\2&1&-2\end{bmatrix}}$

$$AB=\begin{matrix}\overset{\cancel{7}\ 1\ \cancel{-6}}{\cancel{(1\times7)}+\cancel{(2\times-3)}+(0\times2)} & \overset{\cancel{2}\ 0\ \cancel{-2}}{\cancel{(1\times2)}+\cancel{(2\times-1)}+\cancel{(0\times1)}} & \overset{\cancel{-6}\ 0\ \cancel{6}}{\cancel{(1\times-6)}+\cancel{(2\times3)}+(0\times-2)}\\ \overset{0}{\cancel{(0\times7)}+\cancel{(2\times-3)}+\cancel{(3\times2)}} & \overset{\cancel{-2}\ 1\ \cancel{3}}{\cancel{(0\times2)}+\cancel{(2\times-1)}+\cancel{(3\times1)}} & \overset{0}{\cancel{(0\times-6)}+\cancel{(2\times3)}+\cancel{(3\times-2)}}\\ \underset{\cancel{7\ -9\ 2}\ 0}{\cancel{(1\times7)}+\cancel{(3\times-3)}+\cancel{(1\times2)}} & \underset{\cancel{2\ -3\ 1}\ 0}{(1\times2)+\cancel{(3\times-1)}+(1\times1)} & \underset{\cancel{-6\ 9\ -2}\ 1}{\cancel{(1\times-6)}+\cancel{(3\times3)}+\cancel{(1\times-2)}}\end{matrix}$$

$$BA=\begin{matrix}\overset{\cancel{7}\ 1\ \cancel{-6}}{\cancel{(7\times1)}+\cancel{(2\times0)}+\cancel{(-6\times1)}} & \overset{\cancel{14}\ \cancel{4}\ 0\ \cancel{-18}}{\cancel{(7\times2)}+\cancel{(2\times2)}+\cancel{(-6\times3)}} & \overset{\cancel{6}\ 0\ \cancel{-6}}{\cancel{(7\times0)}+\cancel{(2\times3)}+\cancel{(-6\times1)}}\\ \underset{\cancel{-3}\ 0\ \cancel{3}}{\cancel{(-3\times1)}+\cancel{(-1\times0)}+\cancel{(3\times1)}} & \underset{\cancel{-6}\ \cancel{-2}\ 1\ \cancel{9}}{\cancel{(-3\times2)}+\cancel{(-1\times2)}+\cancel{(3\times3)}} & \underset{\cancel{-3}\ 0\ \cancel{3}}{\cancel{(-3\times0)}+\cancel{(-1\times3)}+\cancel{(3\times1)}}\\ \underset{\cancel{2}\ 0\ \cancel{-2}}{\cancel{(2\times1)}+\cancel{(1\times0)}+\cancel{(-2\times1)}} & \underset{\cancel{4\ 0\ -6}\ 0}{\cancel{(2\times2)}+\cancel{(1\times2)}+\cancel{(-2\times3)}} & \underset{1}{\cancel{(2\times0)}+\cancel{(1\times3)}+\cancel{(1\times-2)}}\end{matrix}$$

$\therefore A=B^{-1}, B=A^{-1}$



2.4 Exercises

2) b) $A=\begin{bmatrix}4&1\\3&2\end{bmatrix}$

$$A|1=\begin{array}{cc|cc}\cancel{4}&\cancel{1}&\cancel{1}&\cancel{0}\\ \cancel{3}&\cancel{2}&\cancel{0}&\cancel{1}\end{array}\xrightarrow{-\frac{3}{4}R1}\begin{array}{cc|cc}\cancel{0}&\cancel{\frac{5}{4}}&\cancel{-\frac{3}{4}}&\cancel{1}\end{array}\xrightarrow{\times\frac{4}{5}}\begin{array}{cc|cc}0&1&-\frac{3}{5}&\frac{4}{5}\end{array}$$

$$\begin{array}{cc|cc}4&0&\frac{8}{5}&-\frac{4}{5}\end{array}\ (-R2)\xrightarrow{\times 1/4}\begin{array}{cc|cc}1&0&\frac{2}{5}&-\frac{1}{5}\\0&1&-\frac{3}{5}&\frac{4}{5}\end{array}\therefore A^{-1}=\frac{1}{5}\begin{bmatrix}2&-1\\-3&4\end{bmatrix}$$

d) $A=\begin{bmatrix}1&-1&2\\-5&7&-11\\-2&3&-5\end{bmatrix}$

$$A|1=\begin{array}{ccc|ccc}\cancel{1}&\cancel{-1}&\cancel{2}&\cancel{1}&\cancel{0}&\cancel{0}\\ \cancel{-5}&\cancel{7}&\cancel{-11}&\cancel{0}&\cancel{1}&\cancel{0}\\ 0&2&-1&5&1&0\\ \cancel{-2}&\cancel{3}&\cancel{-5}&\cancel{0}&\cancel{0}&\cancel{1}\\ 0&1&-1&2&0&1\end{array}$$

$+5R_1$, $+2R_1$

$$\xrightarrow{+R2}\begin{array}{ccc|ccc}\cancel{1}&\cancel{0}&\cancel{\frac{3}{2}}&\cancel{\frac{7}{2}}&\cancel{\frac{1}{2}}&\cancel{0}\end{array}\xrightarrow{-\frac{3}{2}R3}\begin{array}{ccc|ccc}1&0&0&2&-1&3\end{array}$$

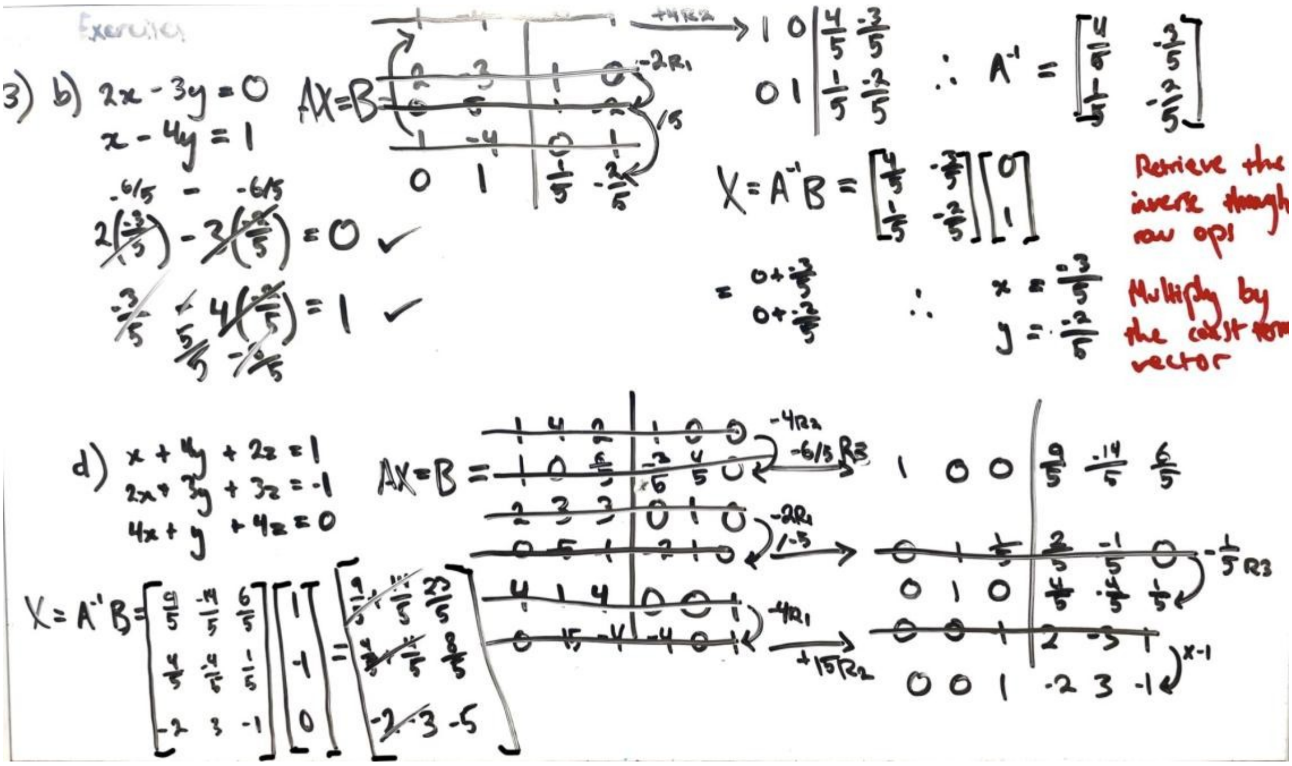
$$\xrightarrow{\times\frac{1}{2}}\begin{array}{ccc|ccc}\cancel{0}&\cancel{1}&\cancel{-\frac{1}{2}}&\cancel{\frac{5}{2}}&\cancel{\frac{1}{2}}&\cancel{0}\end{array}\xrightarrow{+\frac{1}{2}R3}\begin{array}{ccc|ccc}0&1&0&3&1&-1\end{array}$$

$$\xrightarrow{-R2}\begin{array}{ccc|ccc}\cancel{0}&\cancel{0}&\cancel{-\frac{1}{2}}&\cancel{-\frac{1}{2}}&\cancel{-\frac{1}{2}}&\cancel{1}\end{array}\xrightarrow{\times-2}\begin{array}{ccc|ccc}0&0&1&1&1&-2\end{array}$$

$$\therefore A^{-1}=\begin{bmatrix}2&-1&3\\3&1&-1\\1&1&-2\end{bmatrix}$$

Note, we essentially perform row ops to bring A to its identity, the resulting constant terms are A^{-1}

$A|1=1|A^{-1}$



Exercises

3) b) $2x - 3y = 0$
$x - 4y = 1$

$AX = B$

~~$1 \quad -4 \;|\; 0 \quad 1$~~ $\xrightarrow{+4R_2}$ $\left[\begin{array}{cc|cc} 1 & 0 & \frac{4}{5} & \frac{-3}{5} \\ 0 & 1 & \frac{1}{5} & \frac{-2}{5} \end{array}\right]$ $\therefore A^{-1} = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{bmatrix}$

~~$2 \quad -3 \;|\; 1 \quad 0$~~ $-2R_1$

~~$0 \quad 5 \;|\; 1 \quad -2$~~ $/5$

~~$1 \quad -4 \;|\; 0 \quad 1$~~

$0 \quad 1 \;|\; \frac{1}{5} \quad -\frac{2}{5}$

$-6/5 \quad - \quad -6/5$

$2\left(\frac{-3}{5}\right) - 3\left(\frac{-2}{5}\right) = 0$ ✓

$\frac{-3}{5} + \frac{5}{5} - 4\left(\frac{-2}{5}\right) = 1$ ✓ $\quad -\frac{8}{5}$

$X = A^{-1}B = \begin{bmatrix} \frac{4}{5} & -\frac{3}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{bmatrix}\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{matrix} 0 + \frac{-3}{5} \\ 0 + \frac{-2}{5} \end{matrix} \quad \therefore \begin{matrix} x = \frac{-3}{5} \\ y = \frac{-2}{5} \end{matrix}$

Retrieve the inverse through row ops

Multiply by the const term vector

d) $x + 4y + 2z = 1$
$2x + 3y + 3z = -1$
$4x + y + 4z = 0$

$AX = B =$

~~$1 \quad 4 \quad 2 \;|\; 1 \quad 0 \quad 0$~~ $-4R_2$

~~$1 \quad 0 \quad \frac{6}{5} \;|\; \frac{-3}{5} \quad \frac{4}{5} \quad 0$~~ $\xrightarrow{-6/5 R_3}$ $1 \quad 0 \quad 0 \;|\; \frac{9}{5} \quad \frac{-14}{5} \quad \frac{6}{5}$

~~$2 \quad 3 \quad 3 \;|\; 0 \quad 1 \quad 0$~~ $-2R_1$

~~$0 \quad 5 \quad 1 \;|\; -2 \quad 1 \quad 0$~~ $\xrightarrow{/-5}$ ~~$0 \quad 1 \quad \frac{1}{5} \;|\; \frac{2}{5} \quad \frac{-1}{5} \quad 0$~~ $-\frac{1}{5}R_3$

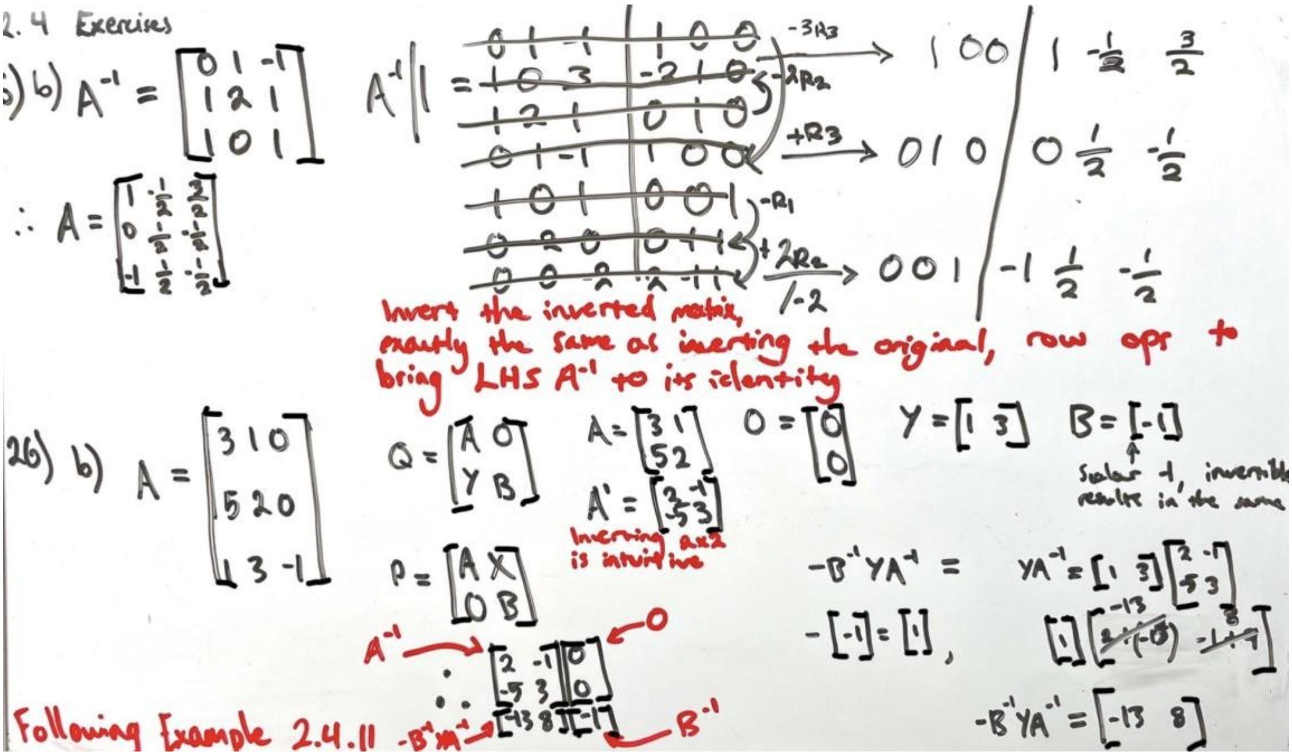
$0 \quad 1 \quad 0 \;|\; \frac{4}{5} \quad \frac{-4}{5} \quad \frac{1}{5}$

~~$4 \quad 1 \quad 4 \;|\; 0 \quad 0 \quad 1$~~ $-4R_1$

~~$0 \quad 15 \quad -4 \;|\; -4 \quad 0 \quad 1$~~ $\xrightarrow[+15R_2]{}$ ~~$0 \quad 0 \quad 1 \;|\; 2 \quad -3 \quad 1$~~ $\times -1$

$0 \quad 0 \quad 1 \;|\; -2 \quad 3 \quad -1$

$X = A^{-1}B = \begin{bmatrix} \frac{9}{5} & \frac{-14}{5} & \frac{6}{5} \\ \frac{4}{5} & \frac{-4}{5} & \frac{1}{5} \\ -2 & 3 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cancel{\frac{9}{5}} + \frac{14}{5} & \frac{23}{5} \\ \cancel{\frac{4}{5}} + \frac{4}{5} & \frac{8}{5} \\ \cancel{-2} - 3 & -5 \end{bmatrix}$



2.4 Exercises

6) b) $A^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$A^{-1} \,|\, I =$

~~$0 \quad 1 \quad -1 \;|\; 1 \quad 0 \quad 0$~~ $-3R_3$

~~$1 \quad 0 \quad 3 \;|\; -2 \quad 1 \quad 0$~~ $-2R_2$ $\xrightarrow{}$ $1 \quad 0 \quad 0 \;|\; 1 \quad -\frac{1}{2} \quad \frac{3}{2}$

~~$1 \quad 2 \quad 1 \;|\; 0 \quad 1 \quad 0$~~

~~$0 \quad 1 \quad -1 \;|\; 1 \quad 0 \quad 0$~~ $\xrightarrow{+R_3}$ $0 \quad 1 \quad 0 \;|\; 0 \quad \frac{1}{2} \quad -\frac{1}{2}$

~~$1 \quad 0 \quad 1 \;|\; 0 \quad 0 \quad 1$~~ $-R_1$

~~$0 \quad 2 \quad 0 \;|\; 0 \quad 1 \quad 1$~~ $+2R_2$

~~$0 \quad 0 \quad 2 \;|\; 2 \quad -1 \quad 1$~~ $\xrightarrow[/-2]{}$ $0 \quad 0 \quad 1 \;|\; -1 \quad \frac{1}{2} \quad -\frac{1}{2}$

$\therefore A = \begin{bmatrix} 1 & -\frac{1}{2} & \frac{3}{2} \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ -1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$

Invert the inverted matrix, exactly the same as inverting the original, row opr to bring LHS A^{-1} to its identity

26) b) $A = \begin{bmatrix} 3 & 1 & 0 \\ 5 & 2 & 0 \\ 1 & 3 & -1 \end{bmatrix}$

$Q = \begin{bmatrix} A & O \\ Y & B \end{bmatrix}$ $\quad A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$ $\quad O = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ $\quad Y = \begin{bmatrix} 1 & 3 \end{bmatrix}$ $\quad B = \begin{bmatrix} -1 \end{bmatrix}$

Scalar -1, invertible results in the same

$A^{-1} = \begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$

Inverting a 2x2 is intuitive

$P = \begin{bmatrix} A & X \\ O & B \end{bmatrix}$

$A^{-1} \rightarrow$ $O \leftarrow$

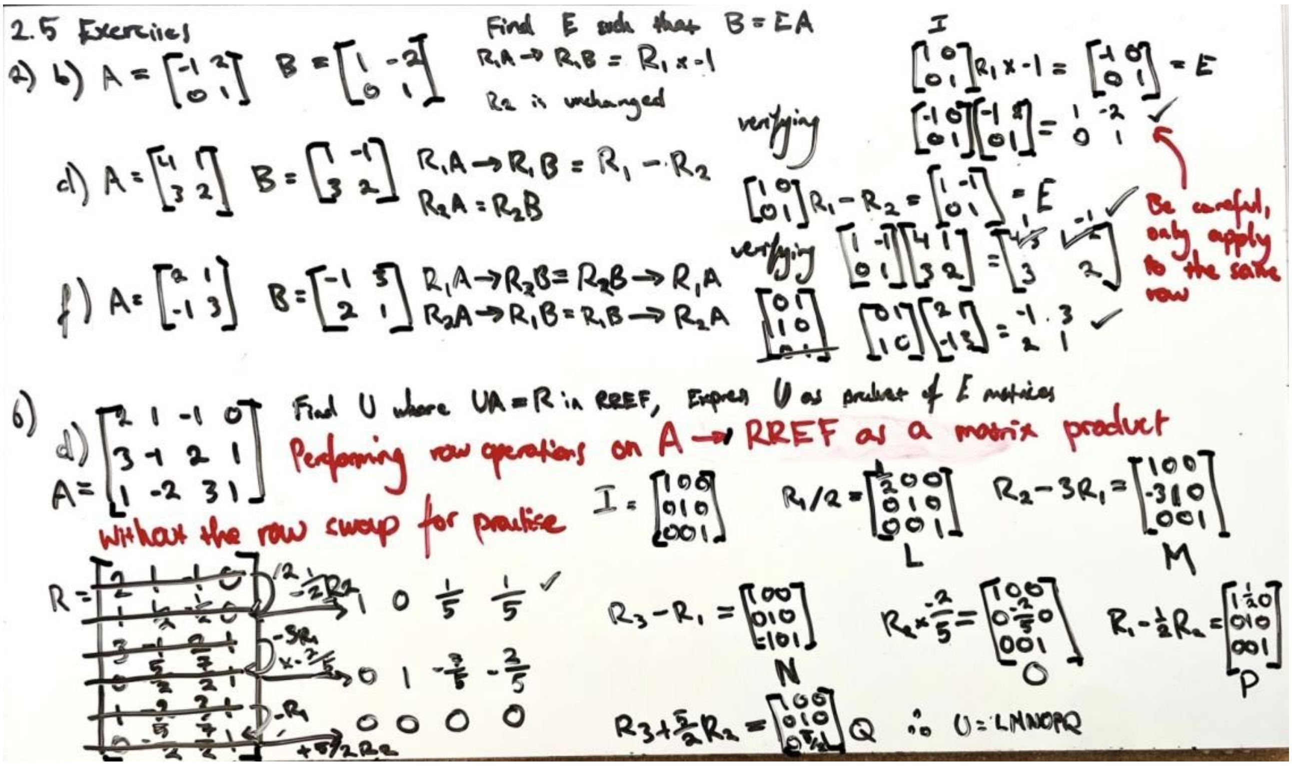
$\therefore \left[\begin{array}{cc|c} 2 & -1 & 0 \\ -5 & 3 & 0 \\ -13 & 8 & -1 \end{array}\right]$

Following Example 2.4.11 $-B^{-1}YA^{-1} \rightarrow$ $\leftarrow B^{-1}$

$-B^{-1}YA^{-1} =$ $\quad YA^{-1} = \begin{bmatrix} 1 & 3 \end{bmatrix}\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$

$-\begin{bmatrix} -1 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix},$ $\quad \begin{bmatrix} 1 \end{bmatrix}\begin{bmatrix} \overset{-13}{\cancel{2+(-15)}} & \overset{8}{\cancel{-1+9}} \end{bmatrix}$

$-B^{-1}YA^{-1} = \begin{bmatrix} -13 & 8 \end{bmatrix}$



2.5 Exercises

Find E such that $B = EA$

2) b) $A = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$ $R_1A \to R_1B = R_1 \times -1$

R_2 is unchanged

$$\overset{I}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} R_1 \times -1 = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} = E$$

verifying $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{matrix} 1 & -2 \\ 0 & 1 \end{matrix}$ ✓

Be careful, only apply to the same row

d) $A = \begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$ $R_1A \to R_1B = R_1 - R_2$

$R_2A = R_2B$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} R_1 - R_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = E$$ ✓

verifying $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} \overset{1}{\cancel{4/3}} & \overset{-1}{\cancel{2}} \\ 3 & 2 \end{bmatrix}$ ✓

f) $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}$ $R_1A \to R_2B = R_2B \to R_1A$

$R_2A \to R_1B = R_1B \to R_2A$

$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} = \begin{matrix} -1 & 3 \\ 2 & 1 \end{matrix}$ ✓

6) d) $A = \begin{bmatrix} 2 & 1 & -1 & 0 \\ 3 & -1 & 2 & 1 \\ 1 & -2 & 3 & 1 \end{bmatrix}$ Find U where $UA = R$ in RREF, Express U as product of E matrices

Performing row operations on $A \to$ RREF as a matrix product without the row swap for practice

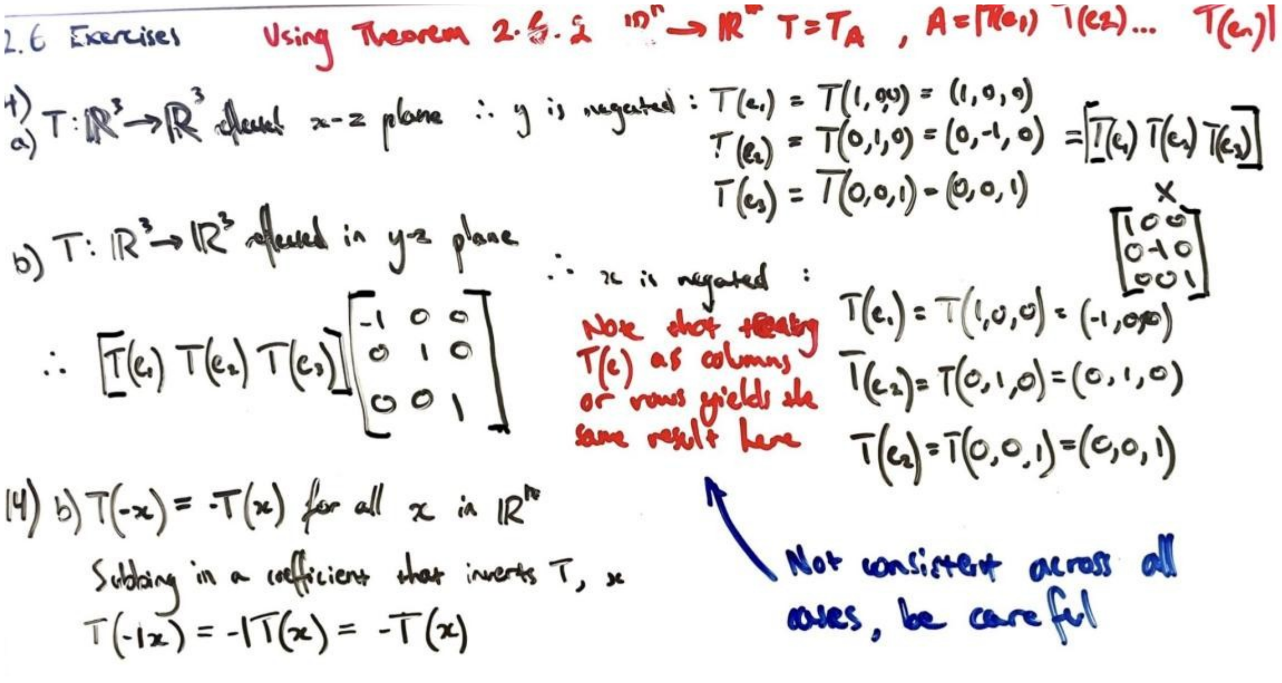
$$R = \begin{matrix} 1 & 0 & \frac{1}{5} & \frac{1}{5} \\ 0 & 1 & -\frac{3}{5} & -\frac{2}{5} \\ 0 & 0 & 0 & 0 \end{matrix}$$ ✓

/2, $-\frac{1}{2}R_2$, $-3R_1$, $\times -\frac{2}{5}$, $-R_1$, $+\frac{5}{2}R_2$

$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $R_1/2 = \underset{L}{\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$ $R_2 - 3R_1 = \underset{M}{\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$

$R_3 - R_1 = \underset{N}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}}$ $R_2 \times \frac{-2}{5} = \underset{O}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{2}{5} & 0 \\ 0 & 0 & 1 \end{bmatrix}}$ $R_1 - \frac{1}{2}R_2 = \underset{P}{\begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}$

$R_3 + \frac{5}{2}R_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5/2 & 1 \end{bmatrix} Q$ $\therefore U = LMNOPQ$



2.6 Exercises

Using Theorem 2.6.2 $\mathbb{R}^n \to \mathbb{R}^m$ $T = T_A$, $A = [T(e_1)\ T(e_2) \dots\ T(e_n)]$

4) a) $T: \mathbb{R}^3 \to \mathbb{R}^3$ reflect x-z plane $\therefore$ y is negated :

$T(e_1) = T(1,0,0) = (1,0,0)$

$T(e_2) = T(0,1,0) = (0,-1,0)$ $= [T(e_1)\ T(e_2)\ T(e_3)]$

$T(e_3) = T(0,0,1) = (0,0,1)$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

b) $T: \mathbb{R}^3 \to \mathbb{R}^3$ reflected in y-z plane $\therefore$ x is negated :

$T(e_1) = T(1,0,0) = (-1,0,0)$

$T(e_2) = T(0,1,0) = (0,1,0)$

$T(e_3) = T(0,0,1) = (0,0,1)$

$$\therefore [T(e_1)\ T(e_2)\ T(e_3)] \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note that treating T(e) as columns or rows yields the same result here

Not consistent across all cases, be careful

14) b) $T(-x) = -T(x)$ for all x in $\mathbb{R}^n$

Subbing in a coefficient that inverts T, x

$T(-1x) = -1T(x) = -T(x)$

3.4.1

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ Find $T\overset{N}{\begin{pmatrix}5\\7\\5\end{pmatrix}}$ if $T\overset{L}{\begin{pmatrix}1\\3\\-1\end{pmatrix}} = \begin{pmatrix}10\\19\\-23\end{pmatrix}$, $T\overset{M}{\begin{pmatrix}1\\-1\\4\end{pmatrix}} = \begin{pmatrix}-6\\-5\\22\end{pmatrix}$

$$a\begin{bmatrix}1\\3\\-1\end{bmatrix} + b\begin{bmatrix}1\\-1\\4\end{bmatrix} = \begin{bmatrix}5\\7\\5\end{bmatrix}$$

$$\left.\begin{matrix}a + b = 5\\ 3a - b = 7\\ -a + 4b = 5\end{matrix}\right\} \cancel{4}a = \frac{\cancel{12}}{\cancel{4}}\ a = 3 \quad \therefore\ b = 2$$

$$\overset{9}{\cancel{3}}(3) - 2 = 7 \checkmark$$

$$-3 + \overset{8}{\cancel{4}}(2) = 5 \checkmark$$

We find constants scalars a, b that take the sum of L, M to N

$$T\begin{bmatrix}5\\7\\5\end{bmatrix} = 3T\begin{bmatrix}1\\3\\-1\end{bmatrix} + 2T\begin{bmatrix}1\\-1\\4\end{bmatrix}$$

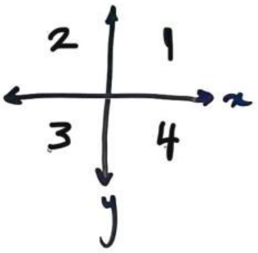
$$= 3\begin{bmatrix}10\\19\\-23\end{bmatrix} + 2\begin{bmatrix}-6\\-5\\22\end{bmatrix}$$

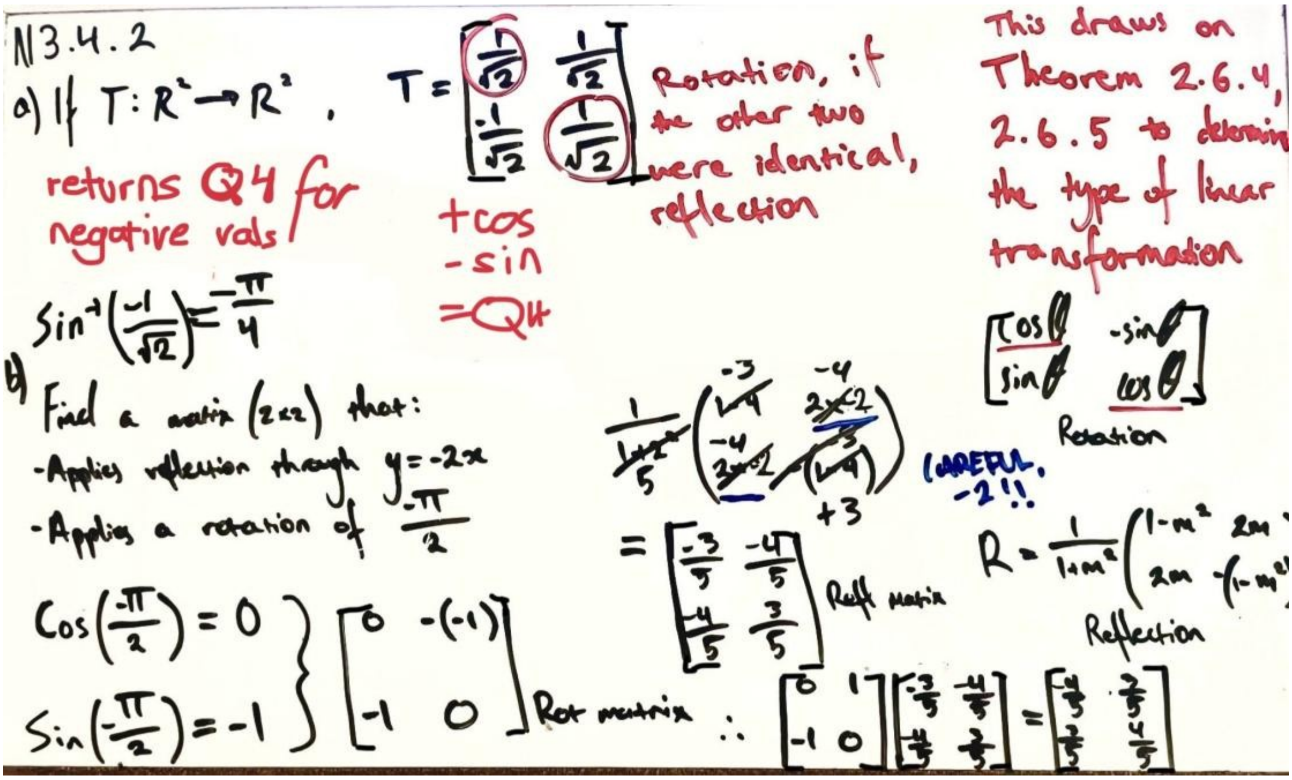
$$= \begin{bmatrix}30\\57\\-69\end{bmatrix} + \begin{bmatrix}-12\\-10\\44\end{bmatrix} = \begin{bmatrix}18\\47\\-25\end{bmatrix}$$

Scale and sum the result, the matrix returned is T(N)

Rotation Table

Q	Angle	Sin θ	Cos θ	Tan θ
1	0-90°	+	+	+
2	90-180°	+	−	−
3	180°-270°	−	−	+
4	270°-360°	−	+	−





N3.4.2

a) If $T: R^2 \rightarrow R^2$, $T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

Rotation, if the other two were identical, reflection

This draws on Theorem 2.6.4, 2.6.5 to determin the type of linear transformation

returns Q4 for negative vals

+cos
-sin
=Q4

$\sin^{-1}\left(\frac{-1}{\sqrt{2}}\right) = -\frac{\pi}{4}$

$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Rotation

b) Find a matrix (2x2) that:

-Applies reflection through $y=-2x$

-Applies a rotation of $\frac{-\pi}{2}$

$\frac{1}{\cancel{1+2^2}\ 5}\begin{pmatrix} \overset{-3}{\cancel{1-4}} & \overset{-4}{\cancel{2\times -2}} \\ \overset{-4}{\cancel{2\times -2}} & \underset{+3}{\cancel{-(1-4)}} \end{pmatrix}$

CAREFUL, -2!!

$= \begin{bmatrix} \frac{-3}{5} & \frac{-4}{5} \\ \frac{-4}{5} & \frac{3}{5} \end{bmatrix}$ Refl matrix

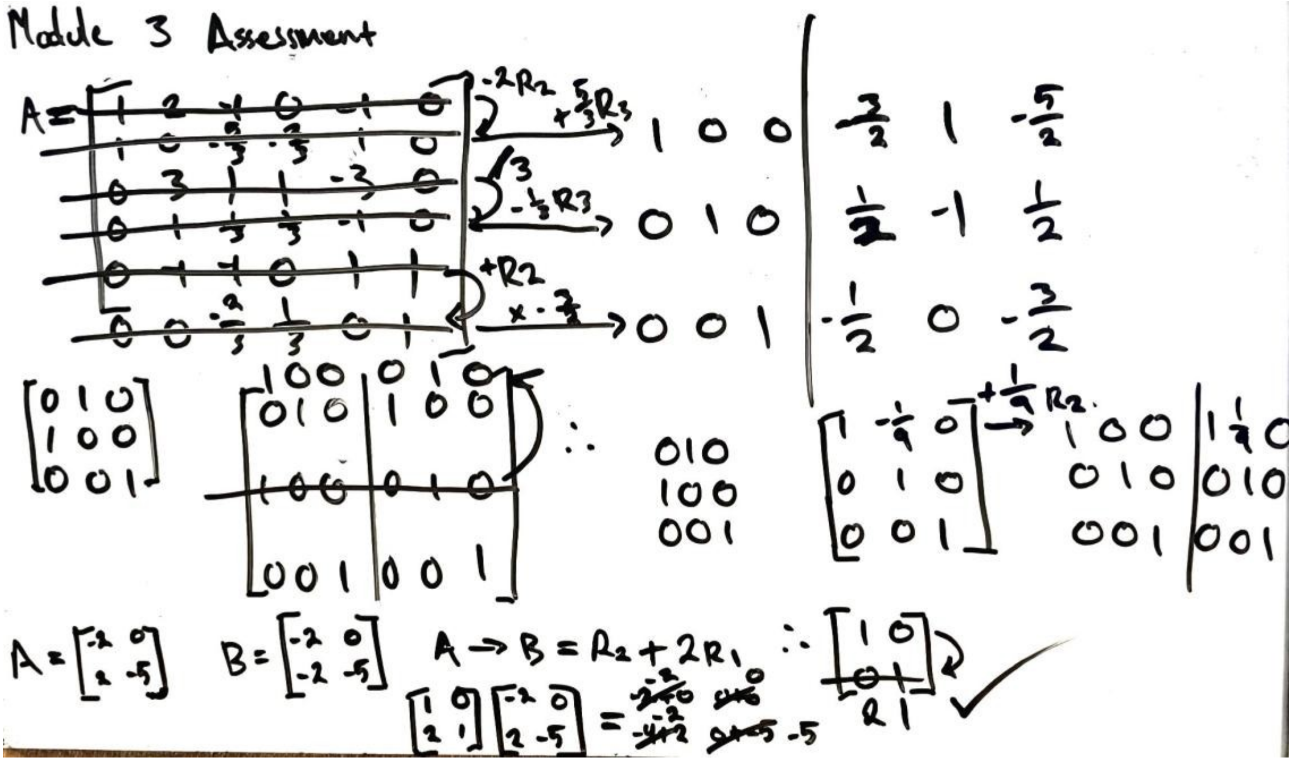
$R = \frac{1}{1+m^2}\begin{pmatrix} 1-m^2 & 2m \\ 2m & -(1-m^2) \end{pmatrix}$

Reflection

$\left.\begin{array}{l} \cos\left(\frac{-\pi}{2}\right) = 0 \\ \sin\left(-\frac{\pi}{2}\right) = -1 \end{array}\right\} \begin{bmatrix} 0 & -(-1) \\ -1 & 0 \end{bmatrix}$ Rot matrix

$\therefore \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} -\frac{3}{5} & -\frac{4}{5} \\ -\frac{4}{5} & \frac{3}{5} \end{bmatrix} = \begin{bmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{3}{5} & \frac{4}{5} \end{bmatrix}$

Attempt 1



Module 3 Assessment

$A = \begin{bmatrix} \cancel{1} & \cancel{2} & \cancel{1} & \cancel{0} & \cancel{-1} & \cancel{0} \\ \cancel{1} & \cancel{0} & \cancel{-\frac{2}{3}} & \cancel{-\frac{2}{3}} & \cancel{1} & \cancel{0} \\ \cancel{0} & \cancel{3} & \cancel{1} & \cancel{1} & \cancel{-3} & \cancel{0} \\ \cancel{0} & \cancel{1} & \cancel{\frac{1}{3}} & \cancel{\frac{1}{3}} & \cancel{-1} & \cancel{0} \\ \cancel{0} & \cancel{1} & \cancel{1} & \cancel{0} & \cancel{1} & \cancel{1} \\ \cancel{0} & \cancel{0} & \cancel{-\frac{2}{3}} & \cancel{\frac{1}{3}} & \cancel{0} & \cancel{1} \end{bmatrix}$

$\xrightarrow[+\frac{5}{3}R_3]{-2R_2} 1\ 0\ 0 \mid -\frac{3}{2}\ \ 1\ \ -\frac{5}{2}$

$\xrightarrow[-\frac{1}{3}R_3]{/3} 0\ 1\ 0 \mid \frac{1}{2}\ \ -1\ \ \frac{1}{2}$

$\xrightarrow[x\cdot -\frac{3}{2}]{+R_2} 0\ 0\ 1 \mid -\frac{1}{2}\ \ 0\ \ -\frac{3}{2}$

$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ \cancel{1} & \cancel{0} & \cancel{0} & \cancel{0} & \cancel{1} & \cancel{0} \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}$

$\therefore$
010
100
001

$\begin{bmatrix} 1 & -\frac{1}{9} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \xrightarrow{+\frac{1}{9}R_2} \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & \frac{1}{9} & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}$

$A = \begin{bmatrix} -2 & 0 \\ 2 & -5 \end{bmatrix}$ $B = \begin{bmatrix} -2 & 0 \\ -2 & -5 \end{bmatrix}$ $A \rightarrow B = R_2 + 2R_1$ $\therefore \begin{bmatrix} 1 & 0 \\ \cancel{0} \; 2 & \cancel{1} \; 1 \end{bmatrix}$ ✓

$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -2 & 0 \\ 2 & -5 \end{bmatrix} = \begin{matrix} \overset{-2}{\cancel{-2+0}} & \overset{0}{\cancel{0+0}} \\ \overset{-2}{\cancel{-4+2}} & \cancel{0+-5}\ -5 \end{matrix}$

$T : \mathbb{R}^2 \to \mathbb{R}^3$ $\quad T\begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-2\\0\\15\end{pmatrix} \quad T\begin{pmatrix}-1\\-1\end{pmatrix} = \begin{pmatrix}2\\2\\9\end{pmatrix}$

Find $T\begin{pmatrix}-2\\1\end{pmatrix}$ $\quad$ find $x, y,$ $\quad x\begin{pmatrix}-2\\-1\end{pmatrix} + y\begin{pmatrix}-1\\-1\end{pmatrix} = \begin{pmatrix}-2\\1\end{pmatrix}$

$-2x - y = -2$

$-x - y = 1$

$\overset{-x}{\cancel{-2x}} - \cancel{y} = \cancel{(-x)} - \cancel{(-y)} = \underset{-3}{\cancel{-2} - 1} \quad \therefore \quad x = 3$

$\cancel{-3} - y = \overset{4}{\cancel{1+3}} \quad \therefore \quad y = -4$

$3\begin{pmatrix}-2\\-1\end{pmatrix} = \begin{pmatrix}-6\\-3\end{pmatrix} = 3\begin{pmatrix}-2\\0\\15\end{pmatrix} = \begin{pmatrix}-6\\0\\45\end{pmatrix}$

$-4\begin{pmatrix}-1\\-1\end{pmatrix} = \begin{pmatrix}4\\4\end{pmatrix} = -4\begin{pmatrix}2\\2\\9\end{pmatrix} = \begin{pmatrix}-8\\-8\\-36\end{pmatrix}$

$+ \quad = \begin{bmatrix}\cancel{-6-8}\\ \cancel{0-8}\\ \cancel{45-36}\end{bmatrix} \begin{matrix}-14\\-8\\9\end{matrix}$

$T : \mathbb{R}^2 \to \mathbb{R}^2 \quad T = \frac{1}{2}\begin{bmatrix}-1 & \sqrt{3}\\ -\sqrt{3} & -1\end{bmatrix} = \begin{bmatrix}\frac{-1}{2} & \frac{\sqrt{3}}{2}\\ -\frac{\sqrt{3}}{2} & -\frac{1}{2}\end{bmatrix}$

$\cos^{-1}\left(\frac{-1}{2}\right) = \frac{2}{3}\pi$